College of Engineering – Department of Electronics Engineering

CURRICULUM

Bachelor of Science in Mechatronics Engineering

Academic Year 2026-2027

Reference CMOs: CMO No. 86, S. 2017, CMO No. 20 s. 2013, CMO No. 39 s. 2021, and CMO No. 40 s. 2021

Curriculum Description

Mechatronics Engineering is an interdisciplinary field that combines mechanical, electrical, electronics, and computer engineering to design and optimize intelligent, automated systems. This integration enables the development of smart technologies that are more efficient, adaptable, and cost-effective than traditional systems. As industries increasingly adopt automation, digital transformation, and smart manufacturing practices, the demand for skilled mechatronics engineers continues to rise—particularly in sectors such as automotive, aerospace, healthcare, agriculture, and semiconductors.

Driven by Industry 4.0 and the Industrial Internet of Things (IIoT), mechatronics engineers are expected to bridge hardware and software systems, supporting innovations in robotics, embedded systems, real-time control, and data-driven automation. This program equips students with technical knowledge, hands-on skills, and industry-aligned experience through practical training, collaborative projects, and partnerships with industry leaders—preparing graduates to meet the evolving challenges of highly automated, technology-driven industrial environments.

Program Education Objectives

The mechatronics engineering alumni three to five years after graduation shall:
PEO 1: Apply knowledge, skills and abilities in mechanical engineering, electrical engineering, electronics engineering and computing in solving inter-disciplinary problems.
PEO 2: Work and lead competently, efficiently and effectively in multi-disciplinary teams to achieve design and/or project objectives.
PEO 3: Participate in lifelong learning to maintain professional, ethical and societal responsibilities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:
SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex mechatronics engineering problems;
SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;
SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.
SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
SO5 Problem Analysis. Identify, formulate, and solve complex mechatronics engineering

problems by applying principles of engineering, science, and mathematics;

SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.

SO7 Communication. Communicate effectively on complex mechatronics engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex mechatronics engineering activities;

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification/ Field / Course	No. of Hours/Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
Differential Calculus	4	0	4
Integral Calculus	4	0	4
Probability and Statistics for AI	3	0	3
Differential Equations	3	0	3
Sub-total	**14**	**0**	**14**
B. Natural/Physical Sciences			
Chemistry for Engineers	3	3	4
Physics for Engineers	3	3	4
Sub-total	**6**	**6**	**8**
C. Basic Engineering Sciences			
Introduction to Engineering	0	3	1

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Engineering Drawing	0	3	1
Computer-Aided Design	0	3	1
Technopreneurship	3	0	3
Materials Science and Engineering	3	0	3
Environmental Science and Engineering	3	0	3
Computer Programming	0	3	1
Basic Occupational Safety and Health	3	0	3
Professional Orientation	2	0	2
Sub-total	**14**	**12**	**18**
D. Allied Courses			
Circuits 1	3	3	4
Engineering Mechanics	3	0	3
Thermodynamics	3	0	3
Electronics Circuits: Devices and Analysis	3	3	4
Circuits 2	3	3	4
Digital Principles and Microcontrollers	3	3	4
Sensors Engineering	1	3	2
Control Systems Engineering	2	3	3
Manufacturing and Quality Control	3	0	3
Industrial Drives and Control	1	3	2
Sub-total	**25**	**21**	**32**
E. Professional Courses			
1. Core Courses			
Advanced Engineering Mathematics	3	0	3
Manufacturing Automation 1	2	3	3
Physical Systems Modelling and Simulation of Machine Elements	2	3	3
Robotics and AI Integration	4	3	5
Manufacturing Automation 2	1	3	2
CAD/CAM and CNC	2	3	3
PLC Fundamentals and Programming	2	3	3
Advanced PLC and Systems Integration	3	3	4

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MexE Capstone Design 1	0	3	1
MexE Capstone Design 2	0	3	1
Supply Chain Management for Mechatronics Engineering	3	0	3
Sub-total	**22**	**27**	**31**
F. INDUSTRY PLACEMENT			
Professional Placement 1	40 hours		8
Professional Placement 2	40 hours		8
Sub-total	**80 hours**		**16**
G. TECHNICAL ELECTIVE			
MexE Elective 1	3	0	3
MexE Elective 2	3	0	3
Sub-total	**6**	**0**	**6**
Total (Technical Courses)	**87**	**66**	**125**
II. NON-TECHNICAL COURSES			
A. General Education Course			
Understanding the Self	3	0	3
Mathematics in the Modern World	3	0	3
The Contemporary World	3	0	3
Readings in Philippine History	3	0	3
Purposive Communication	3	0	3
Ethics	3	0	3
Art Appreciation	3	0	3
Science, Technology and Society	3	0	3
Sub-total	**24**	**0**	**24**
B. Filipino/Literature/Mandated Courses			
Entrepreneurial Mind	3	0	3
Linear Algebra for Artificial Intelligence	3	0	3
Machine Learning and Neural Networks	2	3	3
Life and Works of Rizal	3	0	3
Sub-total	**11**	**3**	**12**
C. Physical Activities Toward Health and Fitness			

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PATHFit 1 - Movement Competency Training	2	0	2
PATHFit 2 - Exercise-based Fitness Activities	2	0	2
PATHFit 3 – Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor & Adventure Activities 1	2	0	2
PATHFit 4 – Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor & Adventure Activities 2)	2	0	2
Sub-total	**8**	**0**	**8**
D. National Service Training Program			
National Service Training Program 1	3	0	3
National Service Training Program 2	3	0	3
Sub-total	**6**	**0**	**6**
E. Institutional Mandated Course			
Foreign Language	**3**	**0**	**3**
Sub-total	**3**	**0**	**3**
Total (Non-Technical Courses)	**52**	**3**	**53**
GRAND TOTAL	**139**	**69**	**178**

CURRICULUM STRUCTURE

FIRST YEAR						
FIRST SEMESTER						
COURSE CODE	**COURSE**	**UNITS**	**NO. OF HOURS**		**PRE REQUISITE**	**CO REQUISITE**
			LEC	**LAB**		
CHEM 101	Chemistry for Engineers	4	3	3		
GEd 102	Mathematics in the Modern World	3	3	0		
GEd 106	Purposive Communication	3	3	0		
GEd 101	Understanding the Self	3	3	0		
MATH 101	Differential Calculus	4	4	0		
NSTP 111	NSTP 1	3	3	0		
PATHFit 1	Movement Competency Training	2	2	0		
ENGG 102	Engineering Drawing	1	0	3		
ENGG 101	****Introduction to Engineering**	1	0	3		
	TOTAL	**24**	**21**	**9**		
FIRST YEAR						
SECOND SEMESTER						

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COURSE CODE	COURSE	UNITS	NO. OF HOURS		PRE REQUISITE	CO REQUISITE
			LEC	LAB		
PHYS 101	Physics for Engineers	4	3	3	MATH 101	
GEd 108	Art Appreciation	3	3	0		
GEd 105	Readings in Philippine History	3	3	0		
GEd 104	The Contemporary World	3	3	0		
MATH 102	Integral Calculus	4	4	0	MATH 101	
NSTP 112	NSTP 2	3	3	0	NSTP 111	
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1	
CpE 101	Computer Programming	1	0	3		
GEd 109	****Science, Technology and Society**	3	3	0		
	TOTAL	**26**	**24**	**6**		
FIRST YEAR						
SUMMER						
COURSE CODE	**COURSE**		**NO. OF HOURS**		**PRE REQUISITE**	**CO REQUISITE**
			LEC	**LAB**		
MATH 103	Differential Equations	3	3	0	MATH 102	
ENGG 118	Materials Science and Engineering	3	3	0	CHEM 101	
IMC 101	Foreign Language	3	3	0		
	TOTAL	**9**	**9**	**0**		

SECOND YEAR						
FIRST SEMESTER						
COURSE CODE	COURSE	UNITS	NO. OF HOURS		PRE REQUISITE	CO REQUISITE
			LEC	LAB		
AI 101	Linear Algebra for Artificial Intelligence	3	3	0	MATH 102	
GEd 107	Ethics	3	3	0		
MexE 204	Advanced Engineering Mathematics for MexE	3	3	0	MATH 103	
EE 207	Circuits 1	4	3	3	MATH 102, PHYS 101	
ENGG 103	Computer-Aided Design	1	0	3	ENGG 102	
ENGG 107	Engineering Mechanics	3	3	0	PHYS 101	
ENGG 113	****Environmental Science and Engineering**	3	3	0	CHEM 101	
ME 101	Thermodynamics	3	3	0	PHYS 101, MATH 102	
ENGG 114	Basic Occupational Safety and Health	3	3	0		

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PATHFit 3	Menu of Dance, Sports, Martial Arts, Group Exercise, Outdoor Adventure Activities Menu Offering: (Traditional and Recreational Games)	2	2	0	PATHFit 2	
	TOTAL	**28**	**26**	**6**		
SECOND YEAR						
SECOND SEMESTER						
COURSE CODE	**COURSE**	**UNITS**	**NO. OF HOURS**		**PRE REQUISITE**	**CO REQUISITE**
			LEC	**LAB**		
ECE 405	Electronics Circuits: Devices and Analysis	4	3	3	EE 207	
EE 208	Circuits 2	4	3	3	EE 207	MexE 204
MexE 206	****Manufacturing Automation 1**	3	2	3	ME 101	
AI 102	ProbabIlity and Statistics for AI	3	3	0	AI 101	
MexE 205	Physical Systems Modelling and Simulation of Machine Elements	3	2	3	ENGG 103, ENGG 107	
PATHFit 4	Menu of Dance, Sports, Martial Arts, Group Exercise, Outdoor Adventure Activities Menu Offering: (Team Sports, Basketball and Volleyball)	2	2	0	PATHFit 2	
MexE 306	Robotics and AI Integration	5	4	3	MexE 204	
ENGG 125	Professional Orientation	2	2	0	2nd yr standing	
	TOTAL	**26**	**21**	**15**		

THIRD YEAR						
FIRST SEMESTER						
COURSE CODE	**COURSE**	**UNITS**	**NO. OF HOURS**		**PRE REQUISITE**	**CO REQUISITE**
			LEC	**LAB**		
ENGG 126	Professional Placement 1	8	640 hours		3rd yr standing, ENGG 125	
	TOTAL	**8**				
THIRD YEAR						
SECOND SEMESTER						
COURSE CODE	**COURSE**	**UNITS**	**NO. OF HOURS**		**PRE REQUISITE**	**CO REQUISITE**
			LEC	**LAB**		
AI 103	Machine Learning and Neural Networks	3	2	3	AI 101, AI 102, CPE 101	
ECE 312	Digital Principles and Microcontrollers	4	3	3	ECE 405	

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ICE 302	Sensors Engineering	2	1	3	3rd yr standing	
MexE 207	****Manufacturing Automation 2**	2	1	3	MexE 206	
MexE 304	CAD/CAM and CNC	3	2	3	MexE 301, ENGG 103	
MexE 305	PLC Fundamentals and Programming	3	2	3	ECE 312, MexE 203	
MexE 403	MexE Capstone Design 1	1	0	3	3rd year standing	
MexEE 401	MexE Elective 1	3	3	0	3rd year standing	
	TOTAL	**21**	**14**	**21**		
THIRD YEAR						
MIDTERM SEMESTER						
COURSE CODE	**COURSE**	**UNITS**	**NO. OF HOURS**		**PRE REQUISITE**	**CO REQUISITE**
			LEC	**LAB**		
ICE 305	Industrial Drives and Control	2	1	3	EE 207, ECE 312	
ECE 409	Control Systems Engineering	3	2	3	MexE 204, EE 208	
GEdE 101	Entrepreneurial Mind	3	3	0		
	TOTAL	**8**	**6**	**6**		

FOURTH YEAR						
FIRST SEMESTER						
COURSE CODE	**COURSE**	**UNITS**	**NO. OF HOURS**		**PRE REQUISITE**	**CO REQUISITE**
			LEC	**LAB**		
MEXE 405	****Supply Chain Management for Mechatronics Engineering**	3	3	0	4th yr standing	
GEd 103	Life and Works of Rizal	3	3	0		
ENGG 105	Technopreneurship	3	3	0	GEdE 101	
IE 406	****Manufacturing and Quality Control**	3	3	0	AI 102	
MexE 401	Advanced PLC and Systems Integration	4	3	3	MexE 305	
MexE 404	MexE Capstone Design 2	1	0	3	MexE 403	
MexEE 402	MexE Elective 2	3	3	0	MexEE 401	

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	TOTAL	**20**	**18**	**6**		
FOURTH YEAR						
SECOND SEMESTER						
COURSE CODE	**COURSE**	**UNITS**	**NO. OF HOURS**		**PRE REQUISITE**	**CO REQUISITE**
			LEC	**LAB**		
ENGG 127	Professional Placement 2	8	640 hours		Graduating	
	TOTAL	**8**	**0**	**0**		

INDUSTRY BASED LEARNING

INDUSTRY -INTEGRATED CURRICULUM FOR BS MECHATRONICS ENGINEERING			
YEAR	**Semester 1**	**Semester 2**	**Midterm**
1	Academic	Academic	Academic
2	Academic	Academic	
3	Industry Placement	Academic	Academic
4	Academic	Industry Placement	**GRADUATE**